MERAQUE

Base Station User Manual

Disclaimer

To ensure the proper and safe operation of your base station, please read and follow all the instructions and safety guidelines outlined in this manual.

Meraque shall not be liable for any damage or loss, whether direct or indirect, legal, special, incidental, or financial (including, but not limited to, loss of income) resulting from improper use or failure to comply with these guidelines. The use of unauthorized components or any modifications that do not align with Meraque's official guidance is strictly discouraged and may void any support or warranty coverage.

Please note that safety protocols and product usage recommendations may be updated periodically. For the most recent version of this manual and other resources, visit www.meraque.ai.

Battery Safety

The Meraque base station is powered by lithium-ion batteries. Improper use of these batteries can be dangerous. Please ensure to follow the battery usage, charging and storage guidelines outlined in this document.

Warning: Only use the battery and charger provided by Meraque. It is prohibited to modify the battery pack and its charger, or use third-party equipment to replace it. The electrolyte in the battery is extremely corrosive. If the electrolyte accidentally splashes into your eyes or skin, please wash the affected area with clean water immediately and seek medical attention.

Precautions

Installation and Setup

- Always operate the base station on frequency bands that comply with your local regulations and laws.
- Place the base station on a flat, level surface for optimal performance and stability.
- Handle cables with care during installation - avoid sharp bends or excessive folding that could damage internal wiring.
- Ensure all antennas are properly positioned with clear line-of-sight and no obstructions.

Operation Precautions

- Operate in open environments with minimal radio interference for best signal quality.
- Before operation, power down other wireless equipment using the same frequency band.
- Only use Meraque-certified accessories and components with your base station - unauthorized accessories may compromise system integrity and create safety hazards.
- Factory-installed components should remain intact - do not attempt to remove or modify them.

Environmental Protection

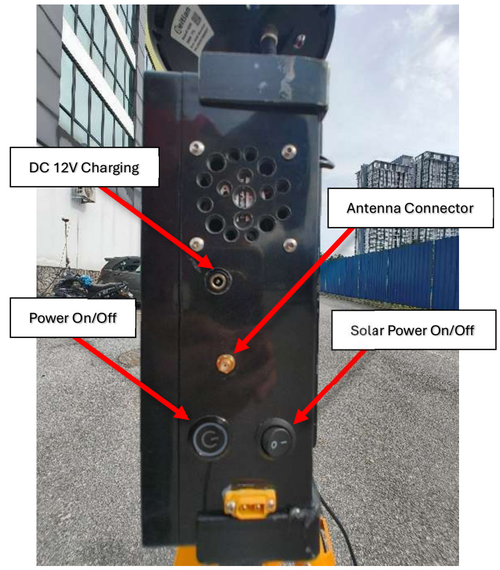
- Protect your base station from contaminants including:
  - All liquids (water, oil, etc.)
  - Sand and dust
  - Other foreign substances
- When operating in rain or snow, use appropriate protective covering to shield the equipment.
- Extreme caution during thunderstorms - suspend operation immediately if lightning is detected in your vicinity.

Following these guidelines will help ensure your base station operates safely and efficiently while maximizing service life and performance.

Introduction

The Meraque base station is a high-precision satellite signal receiver that supports multiple global navigation satellite systems including GPS, Beidou, Galileo, and GLONASS. Engineered with versatility in mind, this advanced RTK base station features an integrated data transmission system specifically designed to operate reliably across diverse applications and deliver centimetre-level positioning accuracy.

Base Station Components



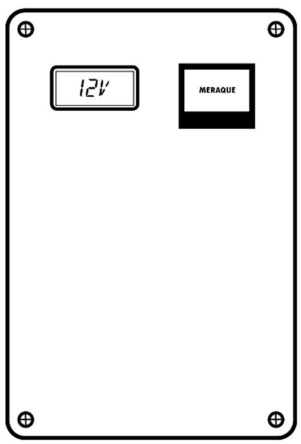
General Preparation for setup

1. **Location:** Choose open area for placing the base station. Once the location is finalized, mark a specific point on the ground where the tripod will be placed. Align the centre of the tripod with this mark to ensure consistent positioning every time the base station is set up, even after relocation.
2. **Prepare the Tripod:** Unfold the tripod completely and extend each of its three legs to the desired height. Once adjusted, securely tighten the top knob to lock the legs in place. Make sure the tripod is firmly placed on the ground and remains stable, as this will provide a solid foundation for the base station.

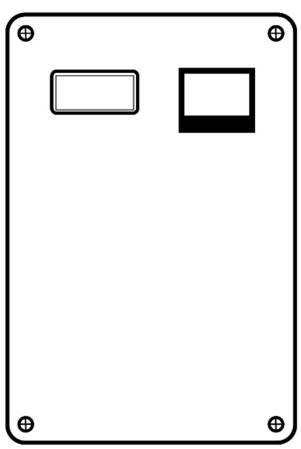
3. **Mount the Base Station:** Attach the main unit of the base station onto the tripod. Secure it tightly

using the lock nut. Verify that the base station is properly levelled and securely mounted.

4. **Antenna Setup:** Attach the antenna wire to the designated port on the base station, ensuring a firm and secure connection.
5. **Turn On/Off:** Press the Power button to turn on the base station. Once the device is power led on, the OLED screen will display "Meraque" to indicate successful startup.



Press the power button to turn off the base station.



Base Station Webpage Setup

Once the base station has been set up at the desired location and powered on, proceed with the following steps to connect your smartphone:

1. On your smartphone, disable mobile data to ensure a stable connection with the base station.
2. Navigate to your phone's Wi-Fi settings. You should see a network named "Base Station" in the list of available networks.
3. Tap on "Base Station" to connect, and when prompted, enter the password: 123456789.
4. Once connected, open any web browser on your smartphone.
5. In the browser's address bar, type or paste the following link: http://192.168.4.1/
6. Wait a few moments for the configuration page to fully load.

You are now connected to the base station interface.

How to use the Base Station

This section provides detailed guidelines for setting up the Real-Time Kinematic (RTK) base station to enhance positional accuracy. The setup process is divided into two operational modes: **Survey Mode** and **Fixed Mode**. Each mode is designed for specific scenarios, ensuring optimal functionality and precision.

Situation:

- **Survey Mode:**
  Use Survey Mode when setting up in a new location or if the base station has been moved slightly, to ensure accurate positioning.

- **Fixed Mode:**
  Use Fixed Mode when the correct location coordinates are already determined, allowing the base station to operate with predefined static values.

Entering GPS Coordinates:

If the Fix-mode is selected, users must input the latitude, longitude, and altitude into the provided fields on the web interface. Adjust the accuracy settings as needed.

- Click 'Update' to apply the coordinates.

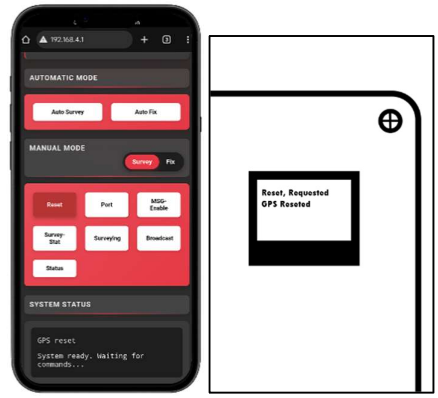
Automatic Setup Modes:

- **Auto Survey:** Use this feature for a quick and automated survey. The base station will automatically gather position data until an accurate reading is obtained.
- **Auto Fix:** This option automatically configures a fixed position using known coordinates, simplifying the setup process when accuracy is critical.

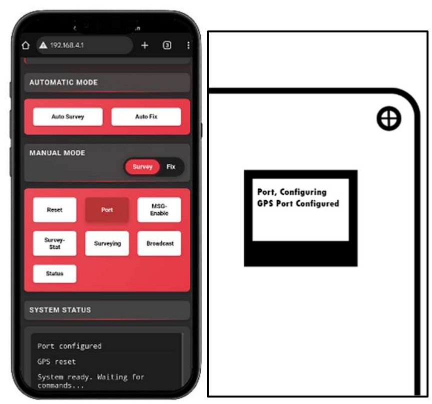
Manual Survey Mode Setup

Use these controls for manual survey operations:

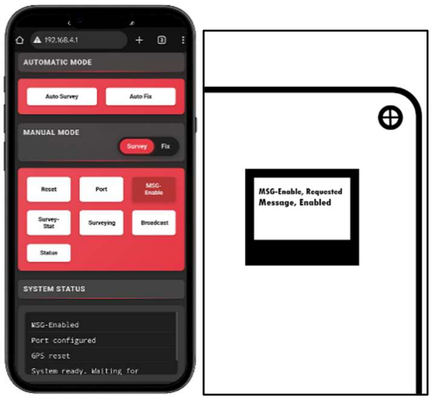
1. **Reset:**
   Press the **"Reset"** button on the webpage. Wait for the display to confirm with the message: *"GPS Reseted."*



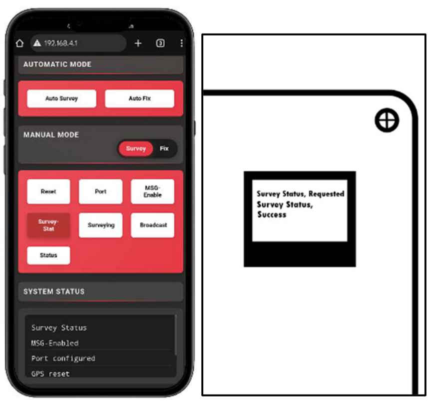
2. **Port Configuration:**
   Press **"Port"** to configure the communication settings to ensure it communicates correctly with other systems.



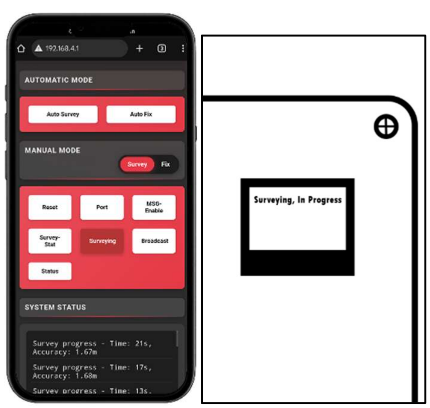
3. **Message Enable:**
   Press **"MSG-Enable"** to set up the device to receive corrections that improve GPS accuracy.

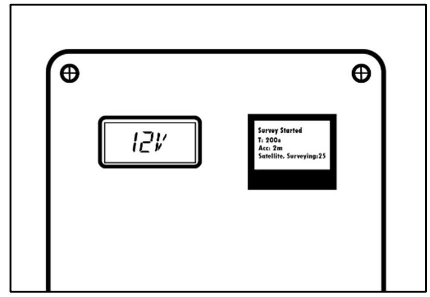


4. **Survey Status:**
   Press **"Survey-Stat"** to verify system readiness.

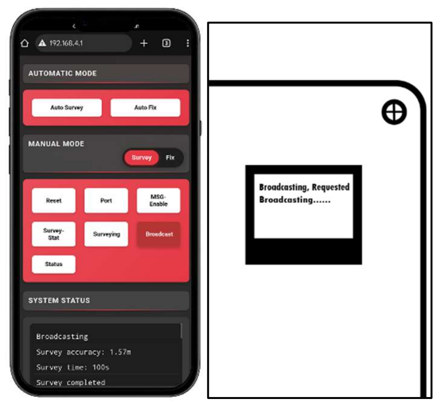


5. **Begin Survey:**
   Press **"Surveying"** and wait for it to complete the survey and find the location.

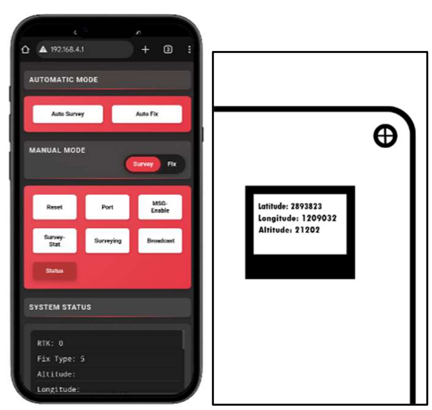




6. **Broadcast Data:**
   Once the survey is complete, press **"Broadcast"** to start transmitting RTK correction data to connected devices.



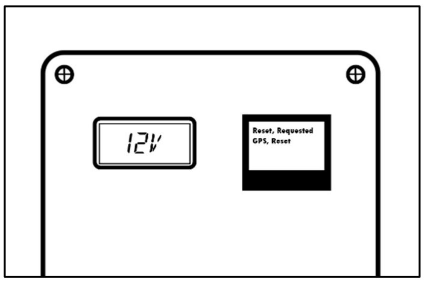
7. **Check Status:**
   Press the **"Status"** button to view the current GPS coordinates (latitude, longitude, and altitude) and confirm successful setup.



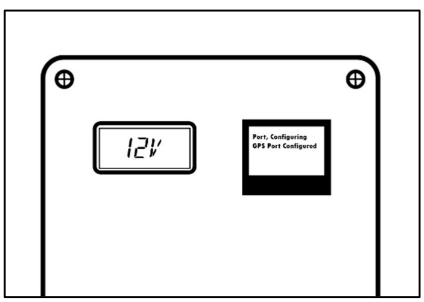
Manual Fixed Mode Setup

For precise positioning control, use these options to manually set up fixed-position operation:

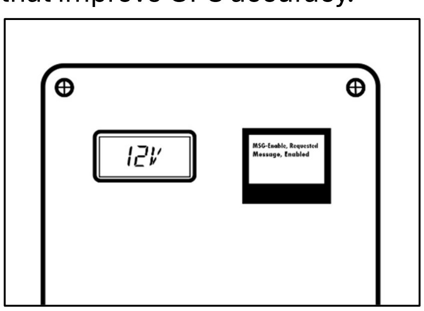
1. **Reset:**
   Press **"Reset"** on the webpage and wait until it says "GPS Reseted" on the display.



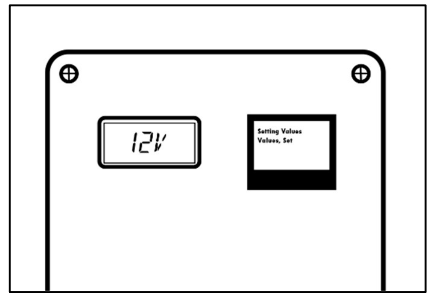
2. **Port:**
   Press **"Port"** to configure the communication settings to ensure it communicates correctly with other systems.



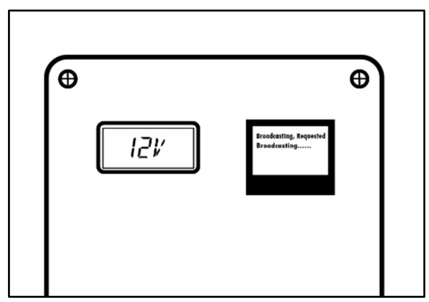
3. **MSG-Enable:**
   Press **"MSG-Enable"** to set up the device to receive corrections that improve GPS accuracy.



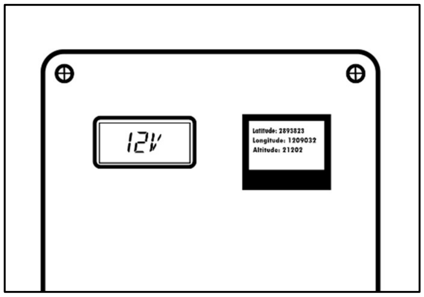
4. **Fixed Mode:**
   At the top of the given input field, type the Latitude, Longitude, and Altitude, and press "Update". Then press **"Fix-Mode"** and wait for it to send the static coordinates.



5. **Broadcast:**
   Once the survey is complete, press **'Broadcast'** to start broadcasting the correction data.



6. **Status:**
   Confirm the setup by pressing the **"Status"** button. It will display the Latitude, Longitude, and Altitude of the current GPS location.



Key Notes

- Once the tripod has been leveled, avoid changing its position or adjusting the angle of the tripod or base station.

- Ensure the surrounding environment is open and free from nearby obstructions such as trees, buildings, or structures that may block or reflect GNSS signals.

- Install the base station at least 200 meters away from any high-power radio sources (e.g., TV towers, microwave transmitters) and at least 50 meters away from high-voltage power lines. This helps minimize electromagnetic interference that could affect GNSS signal quality.

- Do not install the base station near large bodies of water or reflective surfaces, as these can cause signal reflection and multipath errors, leading to inaccurate positioning or reduced performance.

- If you plan to reinstall the base station in the same location in the future, record the precise coordinates of the setup to ensure consistency and reduce setup time.

- Always ensure that the base station, tripod, and antenna connections are tightly secured to prevent accidental shifts due to wind, vibrations, or handling. Use weights or anchors if necessary to stabilize the tripod in windy conditions.

Specifications:

GNSS Receiver Capabilities

- **Supported:**
  - **GPS:** L1, L2
  - **BeiDou:** B1, B2,
  - **GLONASS:** G1, G2
  - **Galileo:** E1, E5b
- **Positioning Accuracy**
  - **Single Point:**
    - Horizontal: 1.5 CEP
    - Vertical: 1.5 CEP
  - **RTK:**
    - Horizontal: 0.01 m + 1 ppm CEP
    - Vertical: 0.01 m + 1 ppm CEP

Communication & Data Transmission

- **LoRa Transceiver**
  - **Module :** LoRa 433 MHz
  - **Range:** Up to 12 km (line-of-sight)
  - **Power Output:** 2W
- **Transmission Format:** RTCM for correction data
- **OLED Display:** Real-time monitoring of GNSS status and system operations

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Electrical Characteristics

- **Primary Power Source:** 12V Li-ion Battery Pack
- **Backup/Charging Source:** Solar Panel (via solar charge controller)
- **Voltage Monitoring:** Integrated voltmeter

- **Cooling System:** 12V fan for thermal regulation

Physical & Environmental

- **Mounting Height (recommended):** 2 meters above ground level
- **Deployment Surface:** Stable and open-sky environment
- **Environmental Protection:** Enclosed for dust and splash resistance